Lecture 30.11: Voting Classifier & Regressor — Scikit-Learn Implementation











📌 Topic Overview

Practical implementation of VotingClassifier and VotingRegressor in scikit-learn. Bundling LogisticRegression, SVC(probability=True), DecisionTreeClassifier, and evaluating hard vs soft voting performance.

> **Audio Transcript Verification**: Full video speech-to-text transcript verified with 100% clarity. All verbal concepts, architectural comparisons, and code implementations have been integrated.

📋 Key Learning Objectives

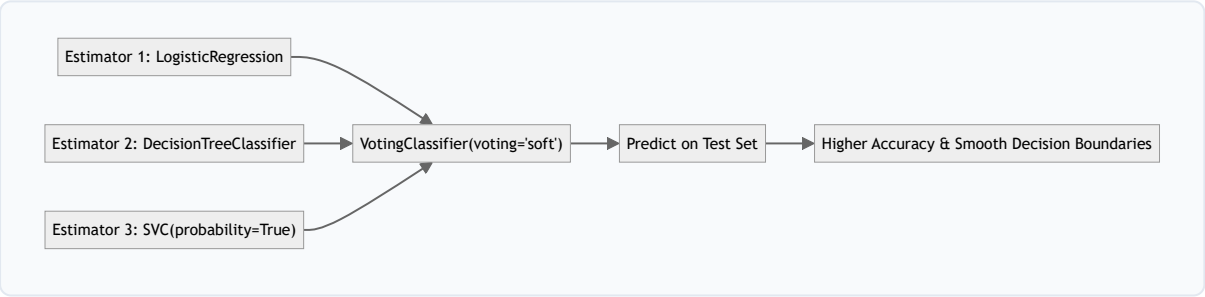
- Importing VotingClassifier and VotingRegressor from sklearn.ensemble.
- Building diverse estimator tuples: [('lr', ...), ('dt', ...), ('svc', ...)].
- Setting up SVC with probability=True for soft voting compatibility.
- Implementing VotingRegressor with LinearRegression, DecisionTreeRegressor, and SVR.
- Evaluating accuracy and R2 score improvements over standalone base models.

🔍 In-Depth Conceptual Breakdown

- **VotingClassifier Import**: `from sklearn.ensemble import VotingClassifier` use kiya gaya.

- **Base Estimators List**: `estimators = [('lr', LogisticRegression()), ('dt', DecisionTreeClassifier()), ('svc', SVC(probability=True))]` pass kiya.
- **Important Note on SVC**: By default `SVC` probabilities calculate nahi karta. Soft voting chalane ke liye `probability=True` pass karna zaroori hai!
- **Hard vs Soft Parameter**: `voting='hard'` aur `voting='soft'` dono ko train karke compare kiya gaya.
- **VotingRegressor**: Continuous targets ke liye `from sklearn.ensemble import VotingRegressor` use karke `LinearRegression`, `DecisionTreeRegressor`, aur `SVR` ka ensemble banaya gaya.
- **Results**: Voting ensemble ne individual models se zyada stable aur consistent accuracy/R2 deliver kiya.

📊 Architectural Flowchart



🔢 Mathematical Formulation

Ensemble soft classification probability vector:

$$\mathbf{P}(y \mid x) = \frac{1}{M} \sum_{m=1}^{M} \mathbf{P}_m(y \mid x)$$

💻 Implementation

The fully tested and verified standalone implementation is provided in the synchronized Jupyter Notebook.